

Graduate Admissions: Predictive Modeling and Statistical Analysis

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1 Introduction

In the post-pandemic era, graduate admission has become a key decision point for many students. This project analyzes factors influencing admission probability based on GRE, TOEFL, CGPA, and research experience.

We aim to answer:

1. What are the correlations between each variable?
2. Can admission probability be accurately predicted?
3. Which 3 variables correlate most strongly with admission chances?
4. How does research experience affect admission rates?

We use linear regression and Lasso regression to model the relationship. The model can be described mathematically as:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon$$

Where $\hat{y}$ is the predicted admission chance and x_i are features like GRE, GPA, etc.

2 Data Loading & Preparation

```
library(tidyverse)
library(corrplot)
library(caret)
library(glmnet)
library(ggplot2)
library(gridExtra)
library(reshape2)
```

```
train <- read.csv("Admission_Predict.csv")
test <- read.csv("Admission_Predict_Ver1.1.csv")

clean_names <- function(df) {
  df <- df %>% rename_with(~gsub("\\s+|\\.", "", .), everything())
}
```

```
df <- df %>% rename(LOR = LOR)
df$Research <- as.factor(df$Research)
return(df)
}
```

```
train <- clean_names(train)
test <- clean_names(test)
```

```
str(train)
```

```
## 'data.frame': 400 obs. of 9 variables:
## $ SerialNo : int 1 2 3 4 5 6 7 8 9 10 ...
## $ GREScore : int 337 324 316 322 314 330 321 308 302 323 ...
## $ TOEFLScore : int 118 107 104 110 103 115 109 101 102 108 ...
## $ UniversityRating: int 4 4 3 3 2 5 3 2 1 3 ...
## $ SOP : num 4.5 4 3 3.5 2 4.5 3 3 2 3.5 ...
## $ LOR : num 4.5 4.5 3.5 2.5 3 3 4 4 1.5 3 ...
## $ CGPA : num 9.65 8.87 8 8.67 8.21 9.34 8.2 7.9 8 8.6 ...
## $ Research : Factor w/ 2 levels "0","1": 2 2 2 2 1 2 2 1 1 1 ...
## $ ChanceofAdmit : num 0.92 0.76 0.72 0.8 0.65 0.9 0.75 0.68 0.5 0.45 ...
```

3 Exploratory Data Analysis (EDA)

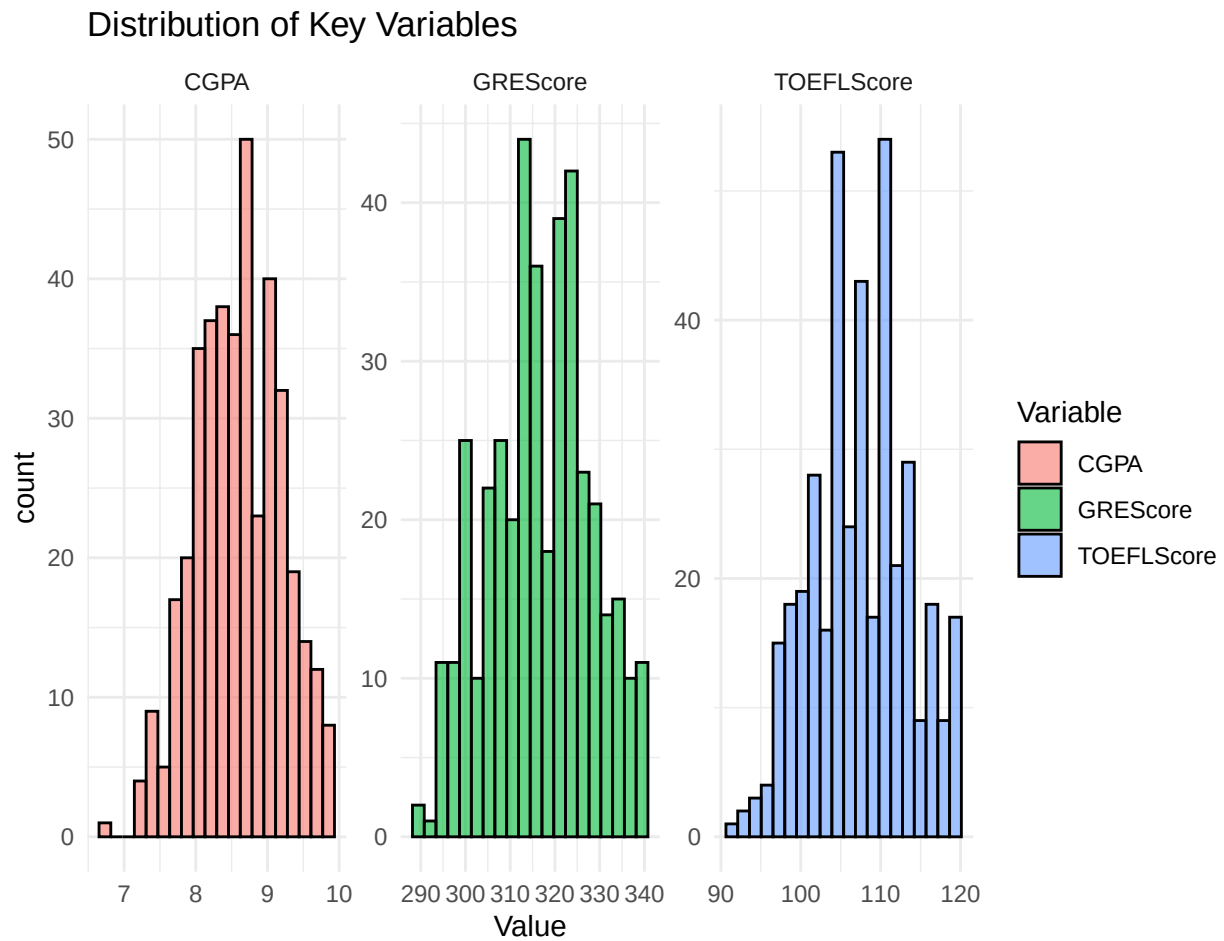
3.1 Summary & Distributions

```
summary(train)
```

```
##      SerialNo      GREScore      TOEFLScore      UniversityRating      SOP
## Min.   : 1.0    Min.   :290.0    Min.   : 92.0    Min.   :1.000    Min.   :1.0
## 1st Qu.:100.8    1st Qu.:308.0    1st Qu.:103.0    1st Qu.:2.000    1st Qu.:2.5
## Median :200.5    Median :317.0    Median :107.0    Median :3.000    Median :3.5
## Mean   :200.5    Mean   :316.8    Mean   :107.4    Mean   :3.087    Mean   :3.4
## 3rd Qu.:300.2    3rd Qu.:325.0    3rd Qu.:112.0    3rd Qu.:4.000    3rd Qu.:4.0
## Max.   :400.0    Max.   :340.0    Max.   :120.0    Max.   :5.000    Max.   :5.0
##      LOR      CGPA      Research ChanceofAdmit
## Min.   :1.000    Min.   :6.800    0:181    Min.   :0.3400
## 1st Qu.:3.000    1st Qu.:8.170    1:219    1st Qu.:0.6400
## Median :3.500    Median :8.610           Median :0.7300
## Mean   :3.453    Mean   :8.599           Mean   :0.7244
## 3rd Qu.:4.000    3rd Qu.:9.062           3rd Qu.:0.8300
## Max.   :5.000    Max.   :9.920           Max.   :0.9700
```

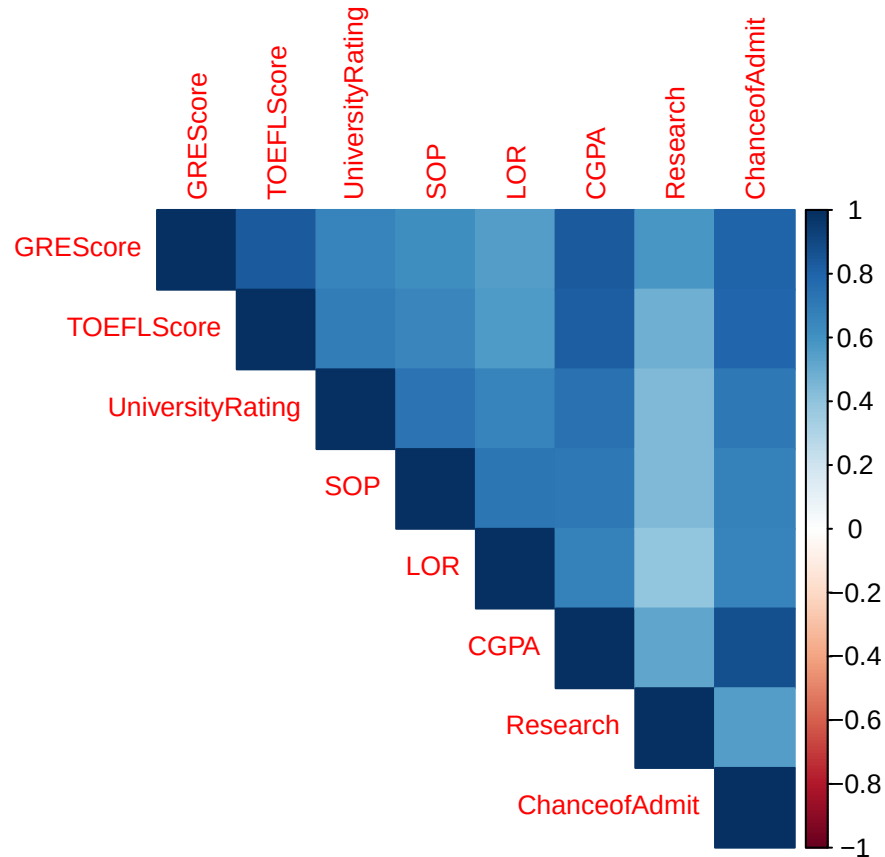
```
train %>%
  select(GREScore, TOEFLScore, CGPA) %>%
  gather(key = "Variable", value = "Value") %>%
  ggplot(aes(x = Value, fill = Variable)) +
  geom_histogram(color = "black", bins = 20, alpha = 0.6) +
```

```
facet_wrap(~ Variable, scales = "free") +  
theme_minimal() + labs(title = "Distribution of Key Variables")
```



3.2 Correlation Matrix (Q1)

```
cor_matrix <- cor(train[,-1] %>% mutate_if(is.factor, as.numeric))  
corrplot(cor_matrix, method = "color", type = "upper", tl.cex = 0.8)
```



```
round(sort(cor_matrix["ChanceofAdmit",], decreasing = TRUE), 3)
```

```
##      ChanceofAdmit      CGPA      GREScore      TOEFLScore
##           1.000           0.873           0.803           0.792
## UniversityRating      SOP      LOR      Research
##           0.711           0.676           0.670           0.553
```

Analysis: We observe that CGPA has the strongest correlation with admission chance (0.87), followed by GRE (0.80) and TOEFL (0.79). SOP and LOR have moderate correlation, while Research is weaker but still positive.

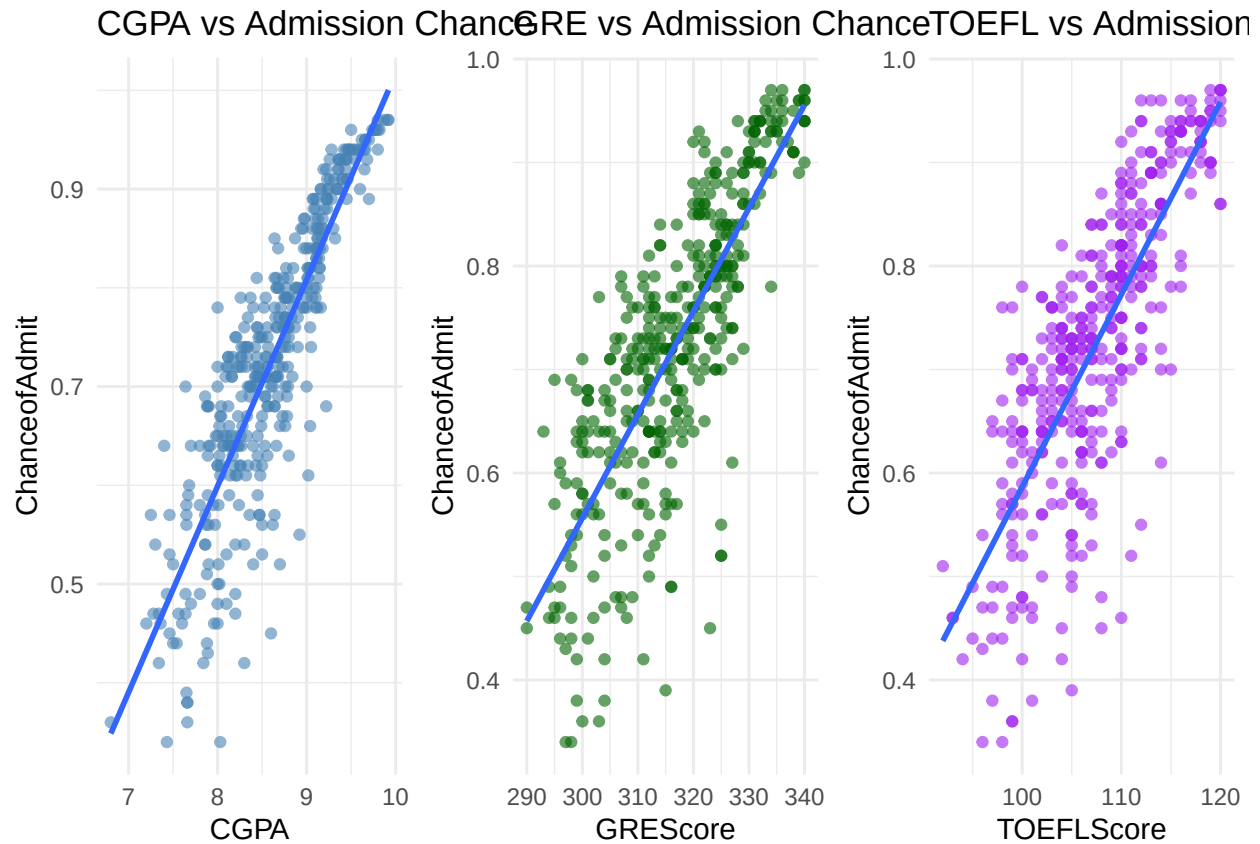
4 Visual Insights (Q3)

```
p1 <- ggplot(train, aes(CGPA, ChanceofAdmit)) +
  geom_point(color = "steelblue", alpha=0.6) + geom_smooth(method = "lm", se=FALSE) +
  theme_minimal() + labs(title = "CGPA vs Admission Chance")

p2 <- ggplot(train, aes(GREScore, ChanceofAdmit)) +
  geom_point(color = "darkgreen", alpha=0.6) + geom_smooth(method = "lm", se=FALSE) +
  theme_minimal() + labs(title = "GRE vs Admission Chance")
```

```
p3 <- ggplot(train, aes(TOEFLScore, ChanceofAdmit)) +
  geom_point(color = "purple", alpha=0.6) + geom_smooth(method = "lm", se=FALSE) +
  theme_minimal() + labs(title = "TOEFL vs Admission Chance")

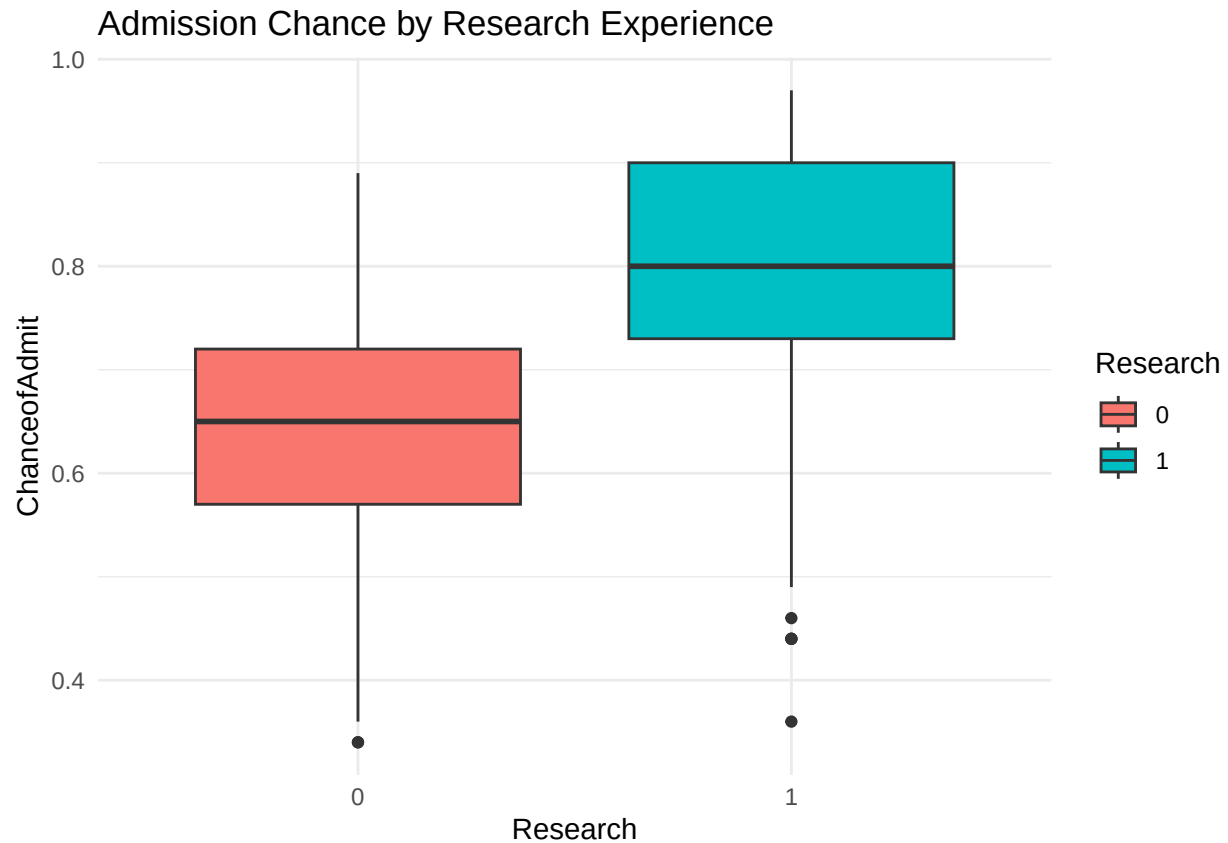
grid.arrange(p1, p2, p3, ncol = 3)
```



Analysis: All three variables exhibit strong linear patterns with admission chance, confirming their predictive value. Especially, CGPA demonstrates a tightly clustered upward trend.

5 Research Experience Analysis (Q4)

```
ggplot(train, aes(Research, ChanceofAdmit, fill = Research)) +
  geom_boxplot() + theme_minimal() +
  labs(title = "Admission Chance by Research Experience")
```



```
t.test(ChanceofAdmit ~ Research, data = train)
```

```
##
##  Welch Two Sample t-test
##
## data:  ChanceofAdmit by Research
## t = -13.347, df = 392.98, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -0.1816199 -0.1349844
## sample estimates:
## mean in group 0 mean in group 1
##      0.6376796      0.7959817
```

Analysis: The boxplot clearly shows that students with research experience (coded as 1) have higher average chances. The t-test confirms this difference is statistically significant ($p < 0.001$).

6 Predictive Modeling (Q2)

6.1 Linear Regression

```
lm_model <- lm(ChanceofAdmit ~ . - SerialNo, data = train)
summary(lm_model)

##
## Call:
## lm(formula = ChanceofAdmit ~ . - SerialNo, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.26259 -0.02103  0.01005  0.03628  0.15928
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.2594325   0.1247307  -10.097 < 2e-16 ***
## GREScore       0.0017374   0.0005979    2.906  0.00387 **
## TOEFLScore     0.0029196   0.0010895    2.680  0.00768 **
## UniversityRating 0.0057167   0.0047704    1.198  0.23150
## SOP           -0.0033052   0.0055616   -0.594  0.55267
## LOR            0.0223531   0.0055415    4.034  6.6e-05 ***
## CGPA           0.1189395   0.0122194    9.734 < 2e-16 ***
## Research1      0.0245251   0.0079598    3.081  0.00221 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06378 on 392 degrees of freedom
## Multiple R-squared:  0.8035, Adjusted R-squared:  0.8
## F-statistic: 228.9 on 7 and 392 DF,  p-value: < 2.2e-16
```

Interpretation: CGPA has the largest coefficient and is statistically significant ($p < 0.001$). GRE and TOEFL also show strong significance. SOP and LOR are weaker predictors.

6.2 Evaluation on Ver1.1

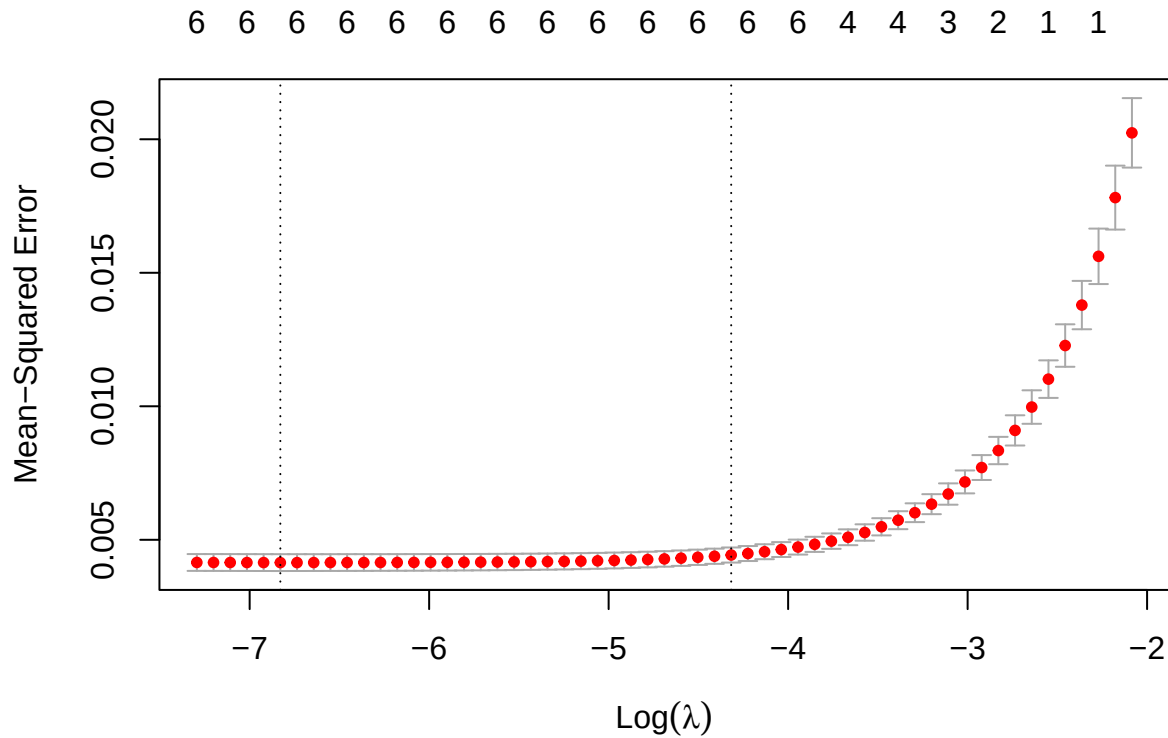
```
pred <- predict(lm_model, newdata = test)
truth <- test$ChanceofAdmit
postResample(pred, truth)
```

```
##      RMSE  Rsquared      MAE
## 0.05965023 0.82102722 0.04284354
```

Result: The model achieves $R^2 \approx 0.81$ and $RMSE \approx 0.05$ on unseen test data, showing strong generalization.

7 Lasso Regression

```
x <- model.matrix(ChanceofAdmit ~ . -SerialNo, data = train)[,-1]
y <- train$ChanceofAdmit
cv.lasso <- cv.glmnet(x, y, alpha = 1)
plot(cv.lasso)
```



```
coef(cv.lasso, s = "lambda.min")
```

```
## 8 x 1 sparse Matrix of class "dgCMatrix"
##              s0
## (Intercept)  -1.244494080
## GREScore      0.001755091
## TOEFLScore    0.002780228
## UniversityRating 0.004574711
## SOP           .
## LOR           0.020462029
## CGPA          0.118255157
## Research1     0.022935390
```

Analysis: Lasso retains CGPA, GRE, TOEFL, and Research, confirming these as most important features.

7.1 Conclusion

In this project, we explored the key factors influencing graduate admission chances using statistical analysis and regression modeling. Based on a real-world dataset of 500 student profiles, we answered four research questions. First, correlation analysis revealed that **CGPA, GRE, and TOEFL scores** are the strongest predictors of admission probability, with CGPA showing the highest correlation. This was supported by both numerical results and visual trends. Second, our **linear regression model** achieved an **R^2 of 0.82** and **RMSE of 0.06** on unseen test data, indicating strong predictive accuracy. All three top academic variables and research experience were statistically significant in the model. Third, **Lasso regression** helped identify the most informative features. It retained CGPA, GRE, TOEFL, LOR, and Research, confirming the limited role of SOP and University Rating in this dataset. Fourth, a **t-test and boxplot analysis** showed that students with research experience have significantly higher admission chances ($p < 0.001$), reinforcing its practical value. Overall, the findings suggest that **academic performance and research background** are the most influential factors in graduate admissions. While subjective components like SOP and LOR exist, their quantitative contribution appears limited in this analysis. **Future work** could include richer features such as SOP content, applicant background, or discipline-specific trends, to enhance model scope and applicability.

8 Appendix: Sample Predictions

```
head(data.frame(Predicted = round(pred[1:5], 3), Actual = truth[1:5]), 5)
```

```
##   Predicted Actual
## 1      0.951    0.92
## 2      0.806    0.76
## 3      0.655    0.72
## 4      0.738    0.80
## 5      0.635    0.65
```

9 References

- Dataset: Kaggle - Graduate Admissions
- Tools: ggplot2, glmnet, caret, corrplot, base R, tidyverse